

ANANT PANDEY

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OVERVIEW

Robotics Engineer specializing in manipulation, simulation, and robot learning. Experienced with ROS 2, MoveIt 2, Isaac Sim and MuJoCo for building robot learning and simulation systems. Experienced across the development lifecycle, from designing robotic systems and collecting data to training models and deploying them on hardware.

SKILLS SUMMARY

- **Programming:** Python, C++, MATLAB, Oracle SQL
- **Platforms:** ROS, ROS 2, Gazebo, Isaac Sim, MuJoCo, MoveIt2
- **Hardware & IoT:** Fusion 360, Raspberry Pi, Arduino, NodeMCU, 3D Printing
- **Data & AI Libraries:** TensorFlow, NumPy, Pandas, LeRobot, Stable-Baselines3
- **Operating Systems:** Windows, Ubuntu

EXPERIENCE

Robotics Engineer

Bangalore

OpenBot

November 2024 – December 2025

- Designed and developed a servo-less leader arm for a leader-follower robotic arm setup, reducing system cost by 90% without compromising functionality or performance.
- Built a robot data collection pipeline using LeRobot, ROS2 and MoveIt2, enabling teleoperation trajectory capture and imitation learning experiments for manipulation tasks.

Freelance Robotics Developer

August 2024 – October 2024

- Developed UR5E simulation environments in Gazebo and Isaac Sim, including IK-based Cartesian teleoperation and wrist camera image capture during trajectory playback.
- Delivered multiple robotics projects, including a custom robotic manipulator, an RRT-based maze solver, and a SLAM-enabled autonomous mobile robot built on Nav2.

Technical Analyst

Bangalore

Deloitte USI

September 2022 – July 2023

- Obtained AI certification from IIT Roorkee, demonstrating expertise in HR technology and AI.
- Contributed to Human Resource Transformation initiatives utilizing Workday and Oracle for performing financial data reconciliation to ensure accuracy and consistency in HR data.

PROJECTS

OpenBot Giraffe

- Designed an affordable, 5-DOF robotic manipulator with a 3D-printed frame and ST3215 servos, targeting hobbyists, educators, and researchers through a modular, open-source platform.
- Integrated with LeRobot, ROS 2, and MoveIt, providing trajectory planning, teleoperation, and imitation based learning capabilities for both simulated and real-world applications.

5-DOF Manipulation Stack

- Engineered a custom 5-DOF IK solver and octomap-based obstacle avoidance for a reduced-DOF arm in Gazebo, planning collision-aware grasps around static obstacles with MoveIt Task Constructor.
- Built a ROS 2 action-server pipeline with multi-object perception and automatic grasp-failure retry for autonomous, repeatable pick-and-place runs.

MuJoCo–Gazebo RL Transfer

- Trained a PPO reach policy from scratch in MuJoCo (Stable-Baselines3), tuning hyperparameters via seed-controlled ablation to raise success rate from 37% to 78%.
- Built a ROS 2 + Gazebo pipeline transferring the policy across simulators, with retry-based trajectory generation and closed-loop hold-and-replan control.

Machine Learning-based NMPC

- Modeled the quadrotor's nonlinear flight dynamics as a state-dependent linear system, and used it in a Nonlinear MPC controller (MATLAB) to replace PID-based angular-rate control
- Trained a neural network (TensorFlow/Keras) on NMPC trajectory data to predict quadrotor state evolution, improving prediction MAE by 47% over the analytical model.

EDUCATION

Manipal Institute of Technology, Manipal

July 2018 – July 2022

B.Tech Electronics and Instrumentation Engineering